

Topic 14: Debugging and Reading Compiler Errors

13.1 Types of Errors

- **Syntax errors:** violate C++ grammar — caught at compile time with line numbers
 - Missing semicolon, mismatched braces, wrong spelling of keywords
- **Runtime errors:** occur while running — program crashes
 - Division by zero, null pointer dereference, array out of bounds
- **Logic errors:** program runs without crashing but gives wrong output
 - Using `=` instead of `==`, off-by-one in loops, wrong formula

13.2 Reading Compiler Error Messages

- Error message format: `filename.cpp:line:column: error: description`
- Start from the first error — later errors are often caused by the first one
- Common messages:
 - `expected ';' before '}'` → missing semicolon on the previous line
 - `undeclared identifier 'x'` → variable not declared or spelled wrong
 - `no matching function for call` → wrong argument types or count
- Fix one error at a time, recompile, and see if others disappear

13.3 Common C++ Errors and Fixes

```
// Error: using = instead of == in condition
if (x = 5)    // assigns 5 to x – always true!
if (x == 5)  // correct comparison

// Error: infinite loop – forgot to increment
int i = 0;
while (i < 5) { cout << i; }    // i never changes
while (i < 5) { cout << i++; } // correct

// Error: array out of bounds
int arr[5];
arr[5] = 10; // index 5 does not exist (0-4 valid)
```

13.4 Debugging with cout

- Insert `cout` statements to print variable values at key points
- `cout << "Before loop, x = " << x << endl;`
- Comment out or remove debug lines after fixing the issue
- Simple but effective — used by professionals too

13.5 Using the VS Code / Visual Studio Debugger

- Set a breakpoint: click to the left of a line number (red dot appears)
- Press F5 to run with debugger — program pauses at the breakpoint
- Step Over (F10): execute the current line and move to the next
- Step Into (F11): go inside a function call
- Watch panel: monitor the value of specific variables as you step
- Call Stack panel: see which functions are currently active

13.6 Debugging Workflow

- Reproduce the bug consistently with a specific input
 - Narrow down which section of code causes it
 - Add breakpoints or `cout` around the suspected area
 - Inspect variable values at that point
 - Fix the code, test again, remove debug output
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